Quiz Questions + Answers

Link to Quiz:
https://docs.google.com/forms/d/e/1FAIpQLSdR0GSYV0Df9CneQCucbFLBDc-TdpdZ2-YrKckfW-c6GnpzbA/viewform?usp=publish-editor

1. Natural image storage
Correct: Logarithmically, with higher sensitivity in shadows. Humans perceive brightness nonlinearly, especially in dark areas, so images are stored with perceptual encoding rather than pure linear values.

- **Linear: Matches physical light, not perception.**
- **Exponential: Incorrect model for image storage.**
- **Both: Sounds reasonable but is not standard encoding practice.**

2. Why is Gamma Correction applied to raw linear camera data?
Correct: To compensate for human logarithmic brightness perception. Camera sensors capture light linearly, but humans do not see brightness linearly.

- **Expand dynamic range: That is HDR.**
- **HDR to LDR: That is tone mapping.**
- **No transformation needed: Ignores human perception differences.**

3. What is the mathematical formula of Gamma Correction
Correct: $s = C \cdot r^{\gamma}$

- **C · log(r): Logarithmic, but not the standard gamma equation.**
- **C / r^γ: Incorrect inversion.**
- **C + γ · r: Only linear scaling.**

4. From the Gamma Correction in the previous equation, what does the 'r' stand for?
Correct: Input pixel value.

- **Output value: That is s.**
- **Scaling constant: That is C.**
- **Shape control: That is γ.**

5. A primary characteristic of HDR images compared to LDR (Typically 8-bit)
Correct: Wider luminance range using 10–12 bits per channel.
HDR preserves more highlight and shadow detail than 8-bit LDR.

- **Less bit depth: Opposite of HDR.**
- **Always tone mapped: False.**
- **Logarithmic capture: Sensors capture light linearly.**

6. What does gamma decoding do?
Correct: Restores linear light data by reversing gamma compression.

- **More shadow values: Describes encoding, not decoding.**
- **HDR compression: Tone mapping.**
- **Log transform: Not standard gamma decoding.**

7. The key conceptual difference between gamma correction and tone mapping is:
Correct: Gamma correction converts linear data into perceptual space, while tone mapping compresses HDR into displayable LDR.

- **Advanced gamma: Oversimplification.**
- **Games vs photography: False distinction.**
- **Same formula: Incorrect.**

8. How does light work?
Correct: Linearly, brightness scales proportionally. Twice the light intensity means twice the photons.

- **Logarithmic: Confuses light with perception.**
- **Exponential: Physically incorrect.**
- **Identical to vision: Human perception is processed biologically.**

9. A typical tone mapping operator applies a non-linear curve that is:
Correct: Mostly linear in dark areas, compressing highlights to avoid clipping.

- **Fully linear: Highlights clip.**
- **Exponentially increasing: Worsens clipping.**
- **Identical to gamma: Different process.**

10. What is Luminance?
Correct: A measure of light intensity per unit area.

- **Human perception: Refers to perceived brightness, not luminance itself.**
- **Total emitted light: That is luminous flux.**
- **Result of gamma correction: Unrelated.**